

2(a)	<i>any three from decrease in sugar</i> 1 <i>idea that</i> changing / transition, from sink to source e.g. leaf becomes the source leaf no longer the sink sugars are (now) being moved away from leaf was a growing area, now translocating (sugars) 2 photosynthesis provides enough sugar / (rate of) photosynthesis increases ; A leaf (now) makes enough sugar I makes own food <i>increase in water to max 2</i> 3 increase in size / more cells / increase in (leaf) surface area, (so proportionate increase in water) ; 4 (more cells that need) water to, maintain turgidity / prevent flaccidity / prevent wilting, (because of transpiration) ; 5 increased transpiration / greater number of stomata ; A increased evaporation (greater internal leaf area) 6 water is, a reactant / needed, for photosynthesis ; <i>(in context of increased photosynthesis, more water needed)</i> 7 AVP ; e.g. maturing leaf growth rate slows / mature leaf no growth (more cells, so more) water needed for, cellular reactions / AW e.g. for hydrolysis of starch (more) water needed for, cell elongation / enlarging vacuoles	3
------	---	---

Question	Answer	Marks																				
2(b)	one mark each correct row ;;; column mark can be used <u>instead of</u> row mark (if higher mark achieved) <table><tr><th>feature</th><th>companion cells</th><th>phloem sieve tube element</th><th>xylem vessel element</th></tr><tr><td>cytoplasm</td><td>✓</td><td>✓</td><td>x</td></tr><tr><td>cell surface membrane</td><td>✓</td><td>✓</td><td>x</td></tr><tr><td>lignified cell wall</td><td>x</td><td>x</td><td>✓</td></tr><tr><td>nucleus</td><td>✓</td><td>x</td><td>x</td></tr></table>	feature	companion cells	phloem sieve tube element	xylem vessel element	cytoplasm	✓	✓	x	cell surface membrane	✓	✓	x	lignified cell wall	x	x	✓	nucleus	✓	x	x	4
feature	companion cells	phloem sieve tube element	xylem vessel element																			
cytoplasm	✓	✓	x																			
cell surface membrane	✓	✓	x																			
lignified cell wall	x	x	✓																			
nucleus	✓	x	x																			